

Project Navi

Bachelor Designer & Developer

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PROJECT NAVI

Feasibility Study

Project Scenarios

Introduction

To assess the feasibility of the project, three different development scenarios were considered: an optimistic scenario, a most likely scenario, and a pessimistic scenario. These scenarios help anticipate possible outcomes, identify potential constraints, and prepare appropriate responses throughout the project lifecycle. They provide a realistic view of how the project may evolve depending on technical, organizational, and financial factors.

Scenario Analysis

The following scenarios illustrate different assumptions regarding the project's progress, resource availability, technical challenges, and overall timeline.

Most likely scenario

Most likely scenario					Likely	
Project Context & Objectives		The project is expected to be completed with minor technical challenges, mainly related to accessibility implementation and GPS integration. Core functionalities will be delivered on schedule with only slight adjustments to the project timeline.			90 % Functional Coverage (Core features completed, optional improvements postponed.)	
Develop an accessible mobile application that assists visually impaired users with navigation and location awareness through voice guidance, GPS services, and an intuitive interface. The objective is to improve users' autonomy while ensuring compliance with accessibility standards.	Internal Factors		External Factors			
	Resource Availability	Project Coordination	Available Resources	Scope Stability	Estimated Duration	
	- Developer available throughout the project - Minor workload fluctuations - Development schedule maintained	- Weekly project planning - Regular milestone reviews - Good task prioritization	- Required software available - Free development tools - Limited additional expenses	- Core requirements remain stable - Minor feature adjustments after testing - MVP scope maintained	7 month Possible delay	
	Project Scope		Reporting & Documentation	Internal Communication	Decision-Making Process	Stakeholder Influence
- Requirements Analysis - UI/UX Design - Mobile Application Development - GPS Navigation Integration - Voice Guidance Integration - Nearby Places Feature - User Testing - Accessibility Validation - Documentation	- Documentation updated regularly - Git version control maintained - Progress reports completed		- Regular communication with supervisor - Feedback integrated throughout development - Clear project documentation	- Supervisor validates milestones - Decisions made within reasonable time - Minor review delays	- Positive feedback from supervisor - Accessibility recommendations considered - Limited change requests	100 € Within planned budget
	Technical Expertise		Technical Infrastructure	Competitive Environment	Regulatory Framework	ROI
	- Learning advanced accessibility features - Minor challenges with GPS and voice APIs	- Reliable development workstation - Android testing device available - Stable development environment	- Existing accessibility applications available - Project differentiated by specific features	- GDPR requirements respected - WCAG accessibility guidelines followed - Android accessibility standards	High social value Improved autonomy for visually impaired users and strong accessibility impact.	

Ideal scenario

Ideal scenario					Ideal
Project Context & Objectives	The project progresses according to the initial schedule without major technical or organizational issues. All planned features are implemented, thoroughly tested, and validated for accessibility.				
Develop a fully accessible mobile application that enables visually impaired users to navigate safely and independently using voice guidance, GPS services, and an intuitive interface. All planned objectives are successfully achieved.	Internal Factors		External Factors		100 % Functional Coverage
	Resource Availability	Project Coordination	Available Resources	Scope Stability	Estimated Duration
	- Full availability throughout the project - No scheduling conflicts - Efficient workload management	- Excellent planning - All milestones achieved on time - Smooth task management	- All required resources available - No unexpected expenses	No changes to project requirements	6 month Possible delay
Project Scope	Reporting & Documentation	Internal Communication	Decision-Making Process	Stakeholder Influence	Estimated Budget
- Requirements Analysis - UI/UX Design - Mobile Application Development - GPS Navigation Integration - Voice Guidance Integration - Nearby Places Feature - User Testing - Accessibility Validation - Documentation (Everything working on time)	- Documentation updated continuously - Complete technical documentation - Well-organized Git repository	- Fast communication - Immediate feedback - Excellent collaboration	- Fast approvals - Immediate validation	- Positive feedback - Very few modification requests	Within planned budget
	Technical Expertise	Technical Infrastructure	Competitive Environment	Regulatory Framework	ROI
	- Full mastery of accessibility standards - GPS and voice APIs integrated successfully	- Reliable hardware - Stable software environment - No technical interruptions	- Limited competition - Innovative solution	- Full GDPR compliance - Full WCAG compliance	Very High Complete solution delivered with maximum accessibility and user satisfaction.

Worst scenario

Worst case scenario					Worst
Project Context & Objectives	Technical difficulties, limited availability, and changing requirements significantly impact the project. The development team focuses on delivering only the essential features before the deadline.				
Develop a simplified version of the application focused on essential accessibility features due to technical and time constraints.	Internal Factors		External Factors		70 % Functional Coverage
	Resource Availability	Project Coordination	Available Resources	Scope Stability	Estimated Duration
	- Limited availability - Frequent interruptions - Development delays	- Several planning changes - Missed milestones - Difficult prioritization	- Budget restrictions - Additional software costs - Limited testing resources	- Significant requirement changes - Features removed from MVP	10 month Possible delay
Project Scope	Reporting & Documentation	Internal Communication	Decision-Making Process	Stakeholder Influence	Estimated Budget
- Requirements Analysis - UI/UX Design - Mobile Application Development - GPS Navigation Integration - Voice Guidance Integration - Nearby Places Feature - User Testing - Accessibility Validation - Documentation (Optional functionality are delayed or canceled)	- Documentation incomplete - Delayed updates - Missing technical details	- Communication delays - Slow feedback - Misunderstandings	- Slow validation process - Delayed approvals	- Numerous change requests - Conflicting feedback	25% above initial estimate
	Technical Expertise	Technical Infrastructure	Competitive Environment	Regulatory Framework	ROI
	- Difficulties implementing accessibility features - GPS integration problems - Voice API compatibility issues	- Hardware limitations - Software compatibility issues - Testing device unavailable at times	- Strong competition - Similar applications released during development	- Accessibility compliance requires additional development time - GDPR adjustments delay delivery	Moderate The application provides essential accessibility features but lacks several planned functionalities.

Ante-Mortem Workshop

An ante-mortem workshop was conducted to imagine that the project had failed before it even started. The objective was to identify possible causes of failure and define preventive actions early in the planning phase. This proactive approach helps reduce project risks and improves decision-making throughout the development process.

Potential failure	Possible cause	Preventive action
AI recognition performs poorly	Insufficient training and testing	Test with diverse datasets and real users
Delayed development	Underestimated workload	Include buffer time and monitor progress weekly
Poor accessibility	Limited user feedback	Conduct usability tests with visually impaired users
Budget exceeded	Unexpected AI service costs	Allocate a contingency budget
Low user adoption	Features do not meet user needs	Collect user feedback during development

Scenario Conclusion

The scenario analysis shows that the project remains feasible under most circumstances. While technical challenges and scheduling risks may affect the development process, these risks can be mitigated through careful planning, continuous testing, and regular project monitoring. The most likely scenario is considered the reference for estimating costs, resources, and project duration in the following sections.

Risk Assessment

Introduction

Following the scenario analysis, a risk assessment was carried out to identify potential threats that could affect the success of the project. Each risk was evaluated based on its probability of occurrence and its potential impact on the project's objectives. Appropriate mitigation measures were also defined to reduce their consequences and improve the overall feasibility of the application.

Risk Identification

The identified risks cover technical, organizational, financial, and security aspects of the project.

Risk	Description	Category
AI recognition errors	The AI may incorrectly identify objects or text.	Technical
Accessibility issues	The interface may not fully meet accessibility requirements.	Technical
Development delays	Some tasks may require more time than expected.	Organizational
Budget overruns	AI services or additional tools increase costs.	Financial
Data privacy	Captured images may contain sensitive personal information.	Security

Risk Prioritization

Risk	Probability	Impact	Priority
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AI recognition errors	High	High	High
Development delays	Medium	High	High
Accessibility issues	Medium	High	High
Budget overruns	Low	Medium	Low
Data privacy	Low	High	Medium

Risks with both high probability and high impact are considered critical and require continuous monitoring throughout the project.

Risk Mitigation Plan

Risk	Measure	Corrective actions
AI recognition errors	Test the AI under different conditions	Improve the model and retrain it if necessary
Accessibility issues	Follow accessibility guidelines and involve visually impaired users in testing	Redesign affected interfaces
Development delays	Monitor progress weekly and prioritize essential features	Adjust the project schedule
Budget overruns	Include a contingency budget	Reduce optional features
Data privacy	Encrypt sensitive data	Apply security updates immediately

Preventive measures aim to reduce the likelihood of risks, while corrective actions help minimize their impact if they occur.

Risk Monitoring

Risks will be reviewed during weekly project meetings. Their probability and impact will be reassessed regularly, and new risks will be documented if they arise. The risk register will be updated throughout the project to ensure that mitigation measures remain effective.

Conclusion

The risk assessment demonstrates that the project involves several manageable risks, mainly related to artificial intelligence, accessibility, and project planning. By implementing preventive measures and continuously monitoring the identified risks, the project team can significantly reduce their impact and increase the likelihood of successful completion.

Economic Feasibility

Since this project is developed as part of an academic programme, labour costs are theoretical estimates based on a standard daily rate. They are intended to evaluate the economic feasibility of the project rather than represent actual development expenses.

Introduction

The purpose of the economic feasibility study is to estimate the overall cost of developing the application and determine whether the project can be completed within a realistic budget. The estimation includes development effort, labour costs, and additional expenses such as software services, hosting, and testing.

Work Breakdown Structure

Task	Estimated duration(days)
Analysis/study	

UI/UX design	
Database design	
Backend development	
Mobile app development	
AI development	
Testing and debugging	
Accessibility implementation	
Documentation	
Total	

The project is estimated to require approximately X working days. The longest phases are mobile development and AI integration, as they represent the application's core functionalities.

Labour Cost Estimation

Task	Days	Cost(€)
Analysis/study		
UI/UX design		
Database design		
Backend development		
Mobile app development		
AI development		
Testing and debugging		
Accessibility implementation		
Documentation		
Total		

Additional Costs

Item	Estimated cost(€)
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Google ML Kit / Vision API	
Cloud hosting	
Domain name	
Development tools	
Testing devices	
Contingency reserve	
Total	

Additional costs include software services, cloud infrastructure, testing equipment, and a contingency reserve to cover unforeseen expenses.

Total Project Budget

Category	Cost(€)
Labour costs	
Additional costs	
Total	

Cost-Benefit Analysis

Conclusion

Based on the estimated workload and associated costs, the project appears to be economically feasible. The projected budget remains reasonable for a mobile application integrating artificial intelligence and accessibility features. Proper financial planning and cost monitoring will help ensure that the project remains within budget throughout its development.

Technical Feasibility

Introduction

The purpose of the technical feasibility assessment is to determine whether the proposed application can be

successfully developed using existing technologies and available technical resources. This analysis evaluates the chosen software architecture, development tools, required skills, and potential technical constraints.

Technology Assessment

Component	Technology	Justification
Mobile App	React Native	Enables cross-platform development for Android and iOS using a single codebase, reducing development time and maintenance costs.
Backend API	Python	Provides a high-performance and scalable backend, well suited for handling API requests and integrating AI services.
Database	PostgreSQL	Reliable relational database suitable for storing users, preferences, and application data.
AI	Python/AWS AI services	Python enables the development and integration of AI models, while AWS provides

		scalable services for object detection, text recognition (OCR), and text-to-speech conversion.
Version Control	Git GitHub	Facilitates collaboration and source code management.

All selected technologies are mature, actively maintained, and widely adopted in the software industry. Their extensive documentation and community support reduce technical risks during development.

Technical Requirements

Requirement	Description
Smartphone camera	Required for object and text recognition
Internet connection	Needed for authentication and cloud services (if applicable).
Mobile operating system	Android (initial version).
Voice synthesis	Converts recognized information into spoken feedback.

Most AI features are designed to operate directly on the device, reducing dependency on an internet connection and improving responsiveness.

Skills Assessment

Required skill	Availability	Action
React Native dev	Partially available	learning
Python dev.	Partially available	learning
DB management	Partially available	learning
AI integration	Partially available	Additional documentation and experimentation required.
Accessibility	Available	Consultation of accessibility guidelines and user testing.

Most of the required technical skills are already available. However, additional learning will be necessary regarding artificial intelligence optimization and accessibility standards to ensure the application meets the needs of visually impaired users.

Technical Constraints

Constraint	Impact	Solution
AI accuracy depends on lighting conditions	Recognition may be less reliable	Test under various environments and improve preprocessing.
Smartphone performance varies	Slower execution on older devices	Optimize image processing and memory usage.
Accessibility requirements		

	Interface must be fully usable without visual interaction	Follow WCAG recommendations and conduct user testing.
Battery consumption	Continuous camera usage may reduce battery life	Optimize camera usage and AI processing.

Technical Feasibility Conclusion

The technical analysis indicates that the project is feasible with current technologies. The selected development tools provide all the necessary functionalities to implement the application's core features, including object recognition, text reading, and voice assistance. Although challenges related to AI performance and accessibility exist, they can be addressed through thorough testing, optimization, and compliance with accessibility best practices.

Organizational Feasibility

Introduction

The purpose of the organizational feasibility assessment is to determine whether the project can be effectively managed within the available time, resources, and organizational structure. This section evaluates project governance, communication methods, planning, and resource availability to ensure the successful completion of the application.

Project Governance

Aspect	Description
Project Manager	Student developer
Stakeholder	Project supervisor
End Users	Visually impaired volunteers participating in usability testing
External Stakeholders	Accessibility organizations (if collaboration is established)
Decision making	Decisions are made by the student in consultation with the project supervisor.
Project reviews	Weekly progress meetings
Communication	Microsoft Teams / Discord / Email
Version Control	GitHub
Task Management	Trello

A clear governance structure ensures that project decisions are taken efficiently and that progress is regularly monitored through meetings with the project supervisor.

Project Planning

Gantt

The project schedule has been organized to allow overlapping development phases where appropriate, reducing the overall development time while maintaining a manageable workload.

Resource Availability

Ressource	Availability
Student developer	Available throughout the project
Project supervisor	Weekly guidance

Development computer	Available
Testing smartphones	Available
Development software	Installed and configured
Cloud services (AWS)	Available during development
Visually impaired testers	Required during usability testing phase
Accessibility organizations	If collaboration is established

The participation of visually impaired users during the testing phase is essential to validate the application's accessibility, usability, and overall effectiveness. Their feedback will be used to improve the interface and AI-assisted features before the final release.

Schedule Feasibility

Based on the estimated workload and the proposed project schedule, the planned deadlines are considered realistic. Regular progress monitoring and weekly reviews will help identify any delays early, allowing adjustments to the planning if necessary. A small time buffer has also been included to accommodate unforeseen technical issues.

Organizational Conclusion

The organizational analysis confirms that the project is feasible within the planned timeframe. The governance structure, communication methods, and project planning provide an effective framework for managing the development process. With regular monitoring and appropriate resource allocation, the project objectives can be achieved within the expected schedule.